A Space Blast That Kept Flashing for 27 Days

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Astronomers have found a space explosion that would not stop. The blast, called GRB 220706A, was first seen on July 6, 2022. Its central engine kept flashing in X-rays for around 27 days.

That beats the record for this kind of flashing, about six days, by around 20 days. Gamma-ray bursts, or GRBs, are some of the most powerful explosions in space.

Long GRBs happen when a huge star collapses. That collapse also causes a **supernova** and makes a black hole. GRB 220706A is an ultra-long GRB, a rare kind whose engine stays active for longer. Scientists still do not know what caused it.

Benjamin Gompertz, of the University of Birmingham in England, said flashing after a GRB is common, but it had always come in the first few days. Flashes at around 27 days show something is still powering the blast, he said.

One idea is a **magnetar**, a **neutron star** with a very strong **magnetic field**. The new **supernova** looks like the one linked to GRB 111209A, another ultra-long GRB from 2011 that was powered by a **magnetar**.

Word Treasure

Astronomers -- Scientists who study stars, planets, and other things in space.

supernova -- A huge explosion that happens when a very large star collapses.

magnetar -- A kind of neutron star with a very strong magnetic field.

neutron star -- A very small, very dense leftover star made mostly of tiny particles called neutrons.

magnetic field -- The invisible area around a magnet where it can pull or push on other things.

Background you need

A gamma-ray burst is a huge explosion in space that gives off very strong bursts of energy.

It can happen when a very large star runs out of fuel and falls inward on itself.

The leftover energy can keep glowing for a while as the explosion fades away.

Break it down

WHO: Astronomers, and Benjamin Gompertz of the University of Birmingham in England.

WHAT: A space explosion, GRB 220706A, whose central engine kept flashing in X-rays for around 27 days.

WHERE: Far away in space, where the blast GRB 220706A happened.

WHY: Because its engine stayed active far longer than usual, showing something was still powering the blast.

Why it matters

It shows you that space can still surprise scientists, just like a mystery you keep solving can turn out longer and stranger than you first thought.

Test what you remember

1. How long did the central engine of GRB 220706A keep flashing in X-rays?

- (A) Around 20 days
- (B) Around 27 days
- (C) In the first few days only
- (D) About six days

2. What kind of gamma-ray burst is GRB 220706A?

- (A) A supernova on its own
- (B) An ultra-long GRB
- (C) A black hole
- (D) A magnetar

3. What happens in a long gamma-ray burst?

- (A) A planet explodes
- (B) A huge star collapses
- (C) A new star is born
- (D) The Moon disappears

4. What does the collapse in a long GRB also cause?

- (A) A supernova and a black hole
- (B) Rain on Earth
- (C) A new galaxy
- (D) A comet

5. What did Benjamin Gompertz say about flashing after a GRB?

- (A) It is common, but it had always come in the first few days
- (B) It stops all other stars from shining
- (C) It only happens once
- (D) It never happens

6. According to the article, what is a magnetar?

- (A) A kind of black hole
- (B) A neutron star with a very strong magnetic field
- (C) A star that has stopped shining
- (D) A spaceship engine

Answer key (try the quiz first!): 1: B 2: B 3: B 4: A 5: A 6: B

Questions to think about

1. Benjamin Gompertz: He said flashing after a GRB is common, but it had always come in the first few days, and flashes at around 27 days show something is still powering the blast.

MY THOUGHTS (at least 20 words):